

Appendix for “Sparse Additive Models for Domain Generalization”

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1 Additional Sensitivity Analysis

In this section, we provide a comprehensive sensitivity analysis of the SpADG framework. These experiments validate the robustness of our variable selection mechanism and justify the hyperparameter configurations used in the main text.

1.1 Choice of the Sparsity Parameter q

Since our framework employs the $\ell_{q,1}$ -norm for regularization, we investigate the impact of the parameter q on the efficacy of feature selection. Specifically, we track the importance trajectories of individual features during the training process for both SpADG_K and SpADG_N . Feature importance is quantified by the $\ell_{q,1}$ -norm of the component weights: $\|\alpha^{(t)}\|_q$ for the kernel variant and $\|\theta_t\|_q$ for the neural variant.

Experimental Setup: For this analysis, we utilize the synthetic dataset with the following configuration: number of domains $N = 30$, samples per domain $n = 32$, total feature dimension $d = 70$, and regularization strength $\lambda = 0.1$. We compare the performance under $q = 1$ and $q = 2$ while keeping other settings constant.

Results and Discussion: Tables 1-2 illustrate the evolution of the two informative features (Feature 1 and 2) relative to the noise features (indices 3–70). We observe that $q = 2$ provides a significantly cleaner and more stable separation between active and inactive variables compared to $q = 1$, particularly for the kernel-based variant (SpADG_K). While $q = 1$ tends to produce either overly conservative or noisy estimates, $q = 2$ effectively suppresses the “max others” (noise features) to near-zero values while steadily increasing the weights of informative features. In SpADG_N , even though the absolute scale of weights is higher under $q = 1$, the relative contrast provided by $q = 2$ ensures a more reliable “screening” effect. This observation aligns with our theoretical results

Table 1: Kernel method: importance of two key features and max/min of others at different epochs (q=1,2)

q	Epoch	feature 1	feature 2	max others	min others
1	200	37.89	41.37	8.13	0.88
1	400	50.04	52.79	8.36	1.88
1	600	55.65	57.88	8.18	2.09
2	200	1.24	1.42	0.24	0.00
2	400	1.84	2.07	0.27	0.00
2	600	2.19	2.29	0.25	0.00

Table 2: Neural network: importance of two key features and max/min of others at different epochs (q=1,2)

q	Epoch	feature 1	feature 2	max others	min others
1	20	0.574	0.583	0.021	0.012
1	40	0.481	0.499	0.005	0.003
1	60	0.469	0.478	0.001	0.001
2	20	0.804	0.758	0.033	0.001
2	40	0.764	0.712	0.048	0.000
2	60	0.753	0.702	0.046	0.000

regarding support recovery, confirming that the structured $\ell_{2,1}$ penalty is more robust in distinguishing genuine cross-domain signals from high-dimensional noise.

1.2 Sensitivity to Kernel and Architecture Configurations

The performance of SpADG is inherently tied to the choice of the component function spaces. We evaluate various combinations of the kernels. The mathematical formulations of the kernels used are:

- **Gaussian (RBF) Kernel:** $k(u, v) = \exp(-\frac{\|u-v\|^2}{2\sigma^2})$.
- **Polynomial Kernel:** $k(u, v) = (\gamma\langle u, v \rangle + r)^p$.
- **Linear Kernel:** $k(u, v) = \langle u, v \rangle$.

Experimental Setup: We set $N = 30, n = 32, d = 50$, and $q = 2$. For SpADG_K , we cross-evaluate combinations of $\kappa \in \{\text{RBF}, \text{Poly}\}$ and $k \in \{\text{RBF}, \text{Linear}\}$. For SpADG_N , we compare the kernel embeddings under RBF and linear kernels, which serve as the weighting terms for the sub-networks.

Results and Discussion: The results demonstrate that the choice of the functional space is critical for capturing cross-domain invariance. For SpADG_K , the combination of a Polynomial kernel for distribution embedding (κ) and a Linear kernel for features ($k_{x, k_{x'}}$) achieves the highest accuracy. This suggests that when marginal distributions are mapped into a sufficiently rich space via κ , linear functional components are effective for additive modeling. Conversely,

Table 3: Performance of SpADG_K with different kernel combinations.

Config (κ, k)	Accuracy	F1 Score	TPR	FPR
RBF / RBF	70.88±4.67	0.65±0.13	0.64±0.30	0.21±0.25
RBF / Lin	88.63±1.88	0.88±0.02	0.89±0.02	0.11±0.04
Poly / RBF	74.00±6.03	0.68±0.13	0.64±0.25	0.16±0.15
Poly / Lin	89.13±1.43	0.89±0.01	0.90±0.01	0.11±0.03

Table 4: Performance of SpADG_N with different kernel embedding.

Config (κ, k)	Accuracy	F1 Score	TPR	FPR
- / RBF	90.69±1.66	0.91±0.02	0.90±0.02	0.08±0.03
- / Lin	53.13±4.40	0.53±0.09	0.55±0.16	0.49±0.13

for SpADG_N, RBF-based sub-networks significantly outperform Linear ones, underscoring the necessity of non-linear shape functions in neural additive architectures to capture complex patterns in domain generalization tasks.

1.3 Impact of Regularization Strength λ

The parameter λ dictates the trade-off between empirical risk minimization and structural sparsity. We systematically evaluate the impact of λ to ensure stable generalization.

Experimental Setup: We evaluate $\lambda \in \{0.001, 0.01, 0.1, 1\}$ using the synthetic and real-world datasets with $N = 30, n = 32, d = 50, q = 2$, and Gaussian kernels.

Table 5: Effect of λ in SpADG_K.

λ	Accuracy	F1 Score	TPR	FPR
0.001	78.69±3.37	0.74±0.08	0.67±0.15	0.09±0.10
0.01	78.19±2.57	0.75±0.07	0.69±0.17	0.12±0.16
0.1	75.13±2.50	0.72±0.07	0.71±0.24	0.20±0.23
1	79.00±5.49	0.78±0.06	0.77±0.06	0.18±0.09

Results and Discussion: Both framework variants exhibit remarkable stability across a wide range of λ . However, performance consistently peaks at $\lambda = 1$ for both SpADG_K and SpADG_N. These empirical results underscore that systematic evaluation and sensitivity analysis are indeed necessary; they not only validate the algorithm’s consistency but also provide a crucial guide for optimizing model performance in real-world applications. Stronger structured sparsity effectively filters out irrelevant high-dimensional features, preventing the model from over-fitting domain-specific noise and promoting the selection of truly robust, cross-domain invariant features. Notably, SpADG_N maintains an accuracy above 89% across all tested scales, demonstrating superior inherent robustness in complex OOD scenarios.

Table 6: Effect of λ in SpADG_N .

λ	Accuracy	F1 Score	TPR	FPR
0.001	89.19 $\pm$ 1.20	0.89 $\pm$ 0.02	0.90 $\pm$ 0.03	0.11 $\pm$ 0.01
0.01	90.00 $\pm$ 1.54	0.90 $\pm$ 0.02	0.90 $\pm$ 0.03	0.10 $\pm$ 0.03
0.1	90.63 $\pm$ 1.40	0.91 $\pm$ 0.02	0.90 $\pm$ 0.03	0.08 $\pm$ 0.01
1	90.88 $\pm$ 2.15	0.91 $\pm$ 0.02	0.91 $\pm$ 0.04	0.09 $\pm$ 0.03

2 Complexity Analysis

To complement our theoretical analysis, we report the empirical running times for the proposed SpADG framework and the compared baselines. This section delineates the training latency observed during our experiments on synthetic, HAR, and Parkinson’s Telemonitoring datasets.

Table 7: Runtime (seconds) of Kernel-Based methods across feature dimensions in synthetic experiment.

Algorithm	d=10	d=30	d=50	d=70
SpADG_K	61.13	186.11	300.74	431.62
MTL	0.08	0.08	0.08	0.08
Pooling	0.03	0.02	0.02	0.02

Table 8: Runtime (seconds) of NN-Based methods across feature dimensions in synthetic experiment.

Algorithm	d=10	d=30	d=50	d=70
SpADG_N	41.53	132.18	195.09	269.87
MLDG	7.68	8.41	7.36	7.35
DANN	0.46	0.48	0.43	0.44
GroupDRO	1.46	1.56	1.38	1.40
EQRM	1.44	1.63	1.43	1.38
IRM	1.42	1.61	1.36	1.35
VREx	0.41	0.50	0.39	0.39
CORAL	14.28	16.48	13.65	13.62
SNAM	0.70	2.76	4.77	7.29
RDM	1.46	1.59	1.28	1.32
GMDG	0.52	0.60	0.50	0.51

By investigating the sensitivity of operational overhead to the feature dimension d , we aim to elucidate the trade-off between the “glass-box” transparency of additive models and their computational requirements. To ensure a fair and consistent comparison, all evaluations were conducted on a consistent computational node equipped with an Intel(R) Core(TM) i7-10700 CPU. All reported durations are averaged over multiple independent runs to minimize statistical

variance.

Table 9: Runtime (seconds) of Kernel-Based methods on real datasets.

Algorithm	HAR	Parkinsons Total	Parkinsons Motor
SpADG _K	5724.69	11.04	10.57
MTL	0.18	0.01	0.01
Pooling	0.03	0.00	0.00

Table 10: Runtime (seconds) of NN-Based methods on real datasets.

Algorithm	HAR	Parkinsons Total	Parkinsons Motor
SpADG _N	1100.89	45.47	44.71
MLDG	3.50	6.99	5.20
DANN	0.31	1.03	0.84
GroupDRO	0.78	1.44	1.18
EQRM	0.99	1.34	1.15
IRM	0.68	1.76	1.36
VREx	0.25	0.70	0.54
CORAL	4.20	13.73	10.47
SNAM	109.91	2.53	2.00
RDM	0.92	0.91	0.70
GMDG	0.37	0.58	0.46